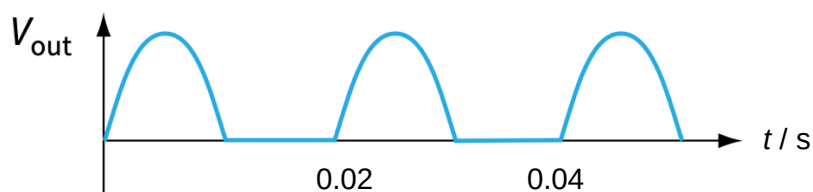


- 6 (a) (i) $100\pi = 2\pi f$, so frequency $f = 50$ Hz.
So period $T = 1/50 = 0.020$ s

B1

(ii)



1 mark for shape
1 mark for labelling of period

B1

B1

- (b) (i) mean power = $\frac{v_{rms}^2}{640} = 40$

B1

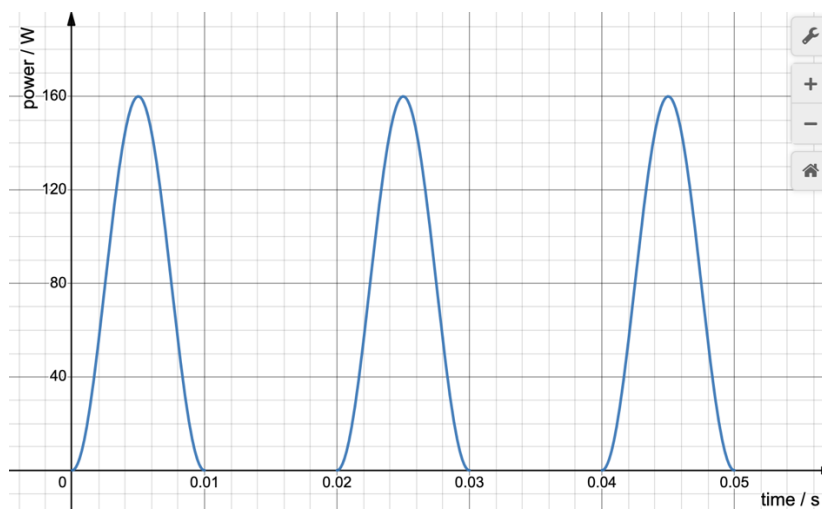
For half-rectified voltage, mean square voltage = $\frac{V_0^2}{4}$

C1

Hence $\frac{V_0^2}{4} = 640 \times 40$, giving $V_0 = 320$ V

A1

(ii)



B1

1 mark for shape
1 mark for labelling of peak power at 160 W

B1

7 (a) (i) photons and electrons undergo a one-to-one interaction B1